THE UNIVERSITY *of* EDINBURGH
School of Physics
and Astronomy

Senior Honours Project

Probing Flow of Complex Fluids Using Optical Tweezers

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Abstract

We use a dual-trap optical tweezers setup to perform microrheology on a simple Newtonian fluid (deionised water) and a viscoelastic complex fluid (PAM solution) probed with 4 μm diameter beads. Trap stiffnesses are obtained from thermal fluctuations via equipartition, confirming that they are an intrinsic property of the trapping setup rather than the surrounding medium. The normalised position autocorrelation function and the normalised bead displacement are used to characterise the viscoelastic response of the complex fluid. Storage and loss moduli are constructed by combining data from stationary and trap flipping measurements, demonstrating that the complex fluid exhibits a stronger viscoelastic response than water.

Declaration

I declare that this project and report is my own work.

Supervisor: Dr. Jochen Arlt 10 Weeks

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1 Introduction

1.1 Complex fluids and rheological background

Complex fluids are a vital part of everyday life, they underpin a wide range of products and processes from personal care, food and biomedicine to hazard assessment [1]. A complex fluid is a material whose flow is governed by an internal evolving microstructure, this includes polymers, colloids, foams, emulsions and biological material [2]. In order to quantify how a material flows or deforms, it is useful for us to introduce some basic rheological quantities. When a material is deformed, we distinguish between stress, which measures the internal force per unit area transmitted through the material, and the strain, which describes how much it is being deformed. In simple shear flow, the shear rate measures how fast layers of the fluid slide past each other [3]. For Newtonian fluids the shear stress is directly proportional to the shear rate, with the constant of proportionality given by the viscosity, therefore the viscosity describes how resistant the fluid is to flow under an applied shear [4, 5].

In complex fluids, the evolving microstructure rearranges, stretches or breaks under shear, and these microscopic changes feed back on the material's macroscopic response, so that the shear stress is no longer simply proportional to the shear rate with a single constant viscosity [2, 5]. Such materials are called non-Newtonian fluids, as they do not follow Newton's law of viscosity, and their viscosity can vary with the applied deformation [6]. The viscosity can depend on shear rate, history and time, leading to shear thinning

where the viscosity decreases as shear rate increases or shear thickening where viscosity increases as shear rate increases [1]. When the imposed shear is removed, the microstructure of many complex fluids can slowly relax towards its original state. This relaxation allows for the material to recover some or all of its initial rheological properties, rather than permanently changing the fluid [7]. Microrheology addresses these behaviours on microscopic lengths by recovering viscoelastic properties from the motion of micron-sized probe particles, which in this work are polystyrene beads [8].

1.2 Optical tweezer microrheology theory

These viscoelastic properties can be obtained through the use of optical tweezers, which provide a controlled microscopic probe to study the response of the fluid. Optical tweezers are a tightly focused laser beam that acts as a vehicle for microrheology by trapping a bead [9]. The strong intensity gradient near the focus of the beam generates a gradient force that pulls the bead towards the region of highest intensity, while the weaker scattering force pushes it back along the beam direction. The balance of these forces creates a stable trapping position close to the focus, allowing the bead to be held and manipulated in the surrounding material [10]. For small displacements from the trapping position, the bead experiences a restoring force that is approximately proportional to its displacement from the trap centre. The trap can therefore be modelled as a harmonic spring with force $\mathbf{F} = -\kappa\,\mathbf{r}$, where κ is the trap stiffness and $\mathbf{r}$ is the displacement of the bead from the trap centre [11]. In this small displacement limit, the bead moves in a harmonic potential of $U(\mathbf{r}) = \frac{1}{2}\,\kappa\,r^2$ [12]. At thermal equilibrium, the probability of finding the bead at a given position is given by Boltzmann's distribution, which for a harmonic potential reduces to a Gaussian [13]. In practice, the bead's position in the image plane can be described by the coordinates x and y, which fluctuate independently about the trap centre at zero. The widths of these fluctuations are set by their variances, $\langle x^2\rangle$ for x and $\langle y^2\rangle$ for y. By the equipartition theorem, each quadratic degree of freedom stores an average potential energy of $\frac{1}{2}k_BT$ where k_B is Boltzmann's constant. This leads to a simple expression for the stiffness along each axis: $\kappa_x = k_BT/\langle x^2\rangle$ and $\kappa_y = k_BT/\langle y^2\rangle$. With the assumption that the trap is approximately isotropic, it is possible to combine both of the directions into a single two-dimensional trap stiffness:

$$\kappa = \frac{2k_BT}{\langle r^2\rangle} \tag{1}$$

where $\langle r^2\rangle = \langle x^2\rangle + \langle y^2\rangle$ is the two-dimensional variance and T is the temperature [10]. This equipartition calibration depends on the thermal fluctuations and the temperature, and therefore the trap stiffness is an intrinsic property of the trapping setup rather than the fluid.

When the trapped bead is suspended in a fluid, its motion results from the balance between the thermal fluctuations and the viscous or viscoelastic drag exerted by the surrounding fluid [14]. The response of the fluid can be described in the frequency domain by the complex shear modulus,

$$G^*(\omega) = G'(\omega) + iG''(\omega). \tag{2}$$

where the storage modulus $G'(\omega)$ quantifies how much elastic energy is stored and released during each cycle of an oscillatory shear deformation at angular frequency ω, and the loss

modulus $G''(\omega)$ quantifies how much energy is dissipated as heat in each cycle [15, 16]. In optical tweezer microrheology the complex modulus is obtained from the trapped bead's motion. A suitable time series is recorded that captures both the low-frequency and high-frequency response of the fluid. At short time scales, thermal fluctuations of a bead held in a single trap probe how quickly the bead can move inside the optical potential, making these fluctuations most sensitive to the high-frequency viscoelastic response of the fluid. This motion is quantified by the normalised position autocorrelation function (NPAF), which describes how rapidly the bead loses memory of its initial position. For a purely Newtonian fluid, the NPAF $A(\tau)$ decays as a single exponential

$$A(\tau) = \exp\left(-\Gamma\tau\right) \tag{3}$$

where τ is the lag time, the time interval between two points in the bead's trajectory whose positions are being compared in the autocorrelation, and Γ is the relaxation rate of the trapped bead given by $\Gamma = \frac{\kappa}{6\pi\eta a}$, where η is the viscosity of the fluid and a is the bead radius [14]. In viscoelastic fluids, the decay of the NPAF is no longer exponential in nature and falls off more slowly or with a different shape than in a Newtonian fluid such as water, reflecting the presence of a relaxation process in the fluid. At longer time scales, a "trap flipping" procedure is used to flip the bead between two different optical traps separated by a fixed distance, and after each switch it relaxes towards the new trap centre. The average relaxation curve reflects the slow, low-frequency response of the surrounding fluid. Combining the information from the single trap thermal fluctuations and the trap flipping experiments, the storage and loss moduli can be reconstructed over a broad range of frequencies to obtain a more complete picture of the fluid's viscoelastic behaviour [14].

To compare different experiments and trap separations, it is convenient to work with a dimensionless, normalised bead displacement, defined as

$$D(t) = \frac{\langle \mathbf{r}(t) \rangle}{D_0}, \tag{4}$$

where $\mathbf{r}(t)$ is the bead's displacement measured from the centre of the active trap and D_0 is the distance between the trap centres, and the angle brackets denote an average over several flips. Immediately after a switch, the bead is displaced by approximately D_0 from the new trap centre, so $D(0) \approx 1$, and as the bead relaxes towards equilibrium $D(t)$ decays towards zero. The shape and timescale of this decay represents the slow, low-frequency viscoelastic response of the fluid [14].

1.3 Aims of this work

The aim of this paper is to validate an optical tweezer microrheology protocol to recover key viscoelastic descriptors, in particular the storage and loss moduli, while also determining the stiffness of the optical trap. The protocol is applied to a simple Newtonian fluid (deionised water) and to a viscoelastic complex fluid composed of polyacrylamide (PAM) solution, in which, flexible polymer chains give rise to pronounced viscoelastic behaviour. Comparing the measured rheological response of water and PAM solution with their expected behaviour allows for the microrheology method to be assessed.

2 Methods

2.1 Apparatus

The experimental setup as shown in Figure 1 consisted of dual-trap optical tweezers built on an inverted microscope (Nikon TE2000M) with a 100x/1.3 oil-immersion objective where 1.3 is the numerical aperture. Two near infrared lasers of wavelength 785 nm generated a fixed and a steerable trap. The steerable beam passed through the beam expander and a pair of computer-controlled galvo mirrors, which were used to move the trap in the sample plane. The fixed beam also passed through the beam expander, but it did not pass through the computer-controlled galvo mirrors so that the trap remained stationary. Both of the beams were then reflected by the dichroic mirror that sends the beams into the objective to form two traps in the sample region, which was illuminated by a light source. The sample was placed coverslip down on top of the objective which was immersed with oil in order to increase the numerical aperture which allows us to focus the laser more tightly and thus achieve a stronger trap. The sample was then imaged and recorded with a camera (XiQ MQ013MG-ON) mounted on the microscope's camera port and connected to a PC running an in-house built programme to control both of the traps, together with Micro-Manager to record videos.

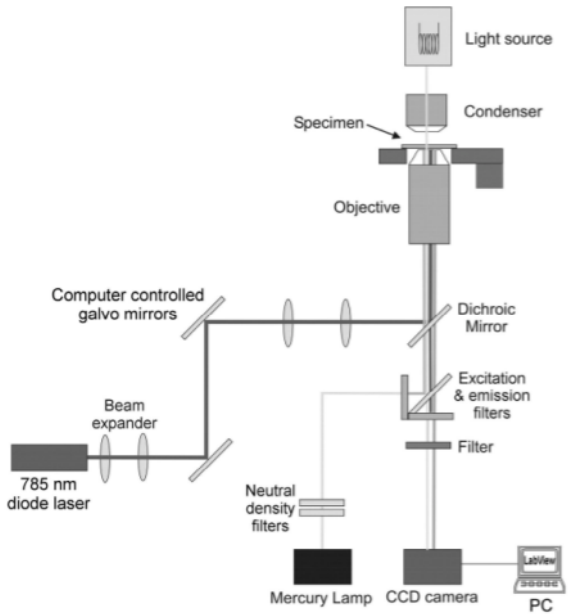


Figure 1: Schematic of the optical-tweezers setup, featuring a 785 nm beam which is expanded and controlled by galvo mirrors, reflected by a dichroic mirror into a 100x objective to form the traps, and the sample is then imaged onto a camera. Courtesy of Dr. Jochen Arlt.

2.2 Experimental Procedure

2.2.1 Sample preparation

To begin the experiment, 4 μm diameter beads were dispersed in deionised water at a concentration of 0.01% (v/v). This dilution was chosen so that the beads were still easy to find and sufficiently far apart to avoid bead-bead interactions that could disturb the traps whilst making measurements. As shown in Figure 2 the sample was built on top of a glass slide with a shallow enclosure made with double-sided tape to host the colloidal solution. This was then enclosed with a coverslip in order to seal the edges and reduce evaporation, any potential leaks and drift in the sample. The coverslip was then placed in contact with the immersive oil on the objective, providing index matching for tighter, stronger and more stable traps [17]. This simple design allowed clear imaging with a large area to find isolated beads. Importantly, before any measurements were made, the temperature of the room was recorded with the laboratory thermostat before each data acquisition. Additionally, the sample was left to settle for around 3 minutes so that the beads could disperse evenly and to further alleviate the drift in the solution. For the viscoelastic measurements, an additional sample was prepared using the PAM solution mixed with deionised water at a concentration of 0.1% (w/w). The same 4 μm beads were dispersed at a concentration of 0.01% (v/v), so that the bead density and bead-bead interactions were comparable to the water sample while the surrounding fluid properties were changed. The PAM solution was placed in an identical tape enclosure and sealed with a coverslip using the same procedure as for the water sample. Similarly to the water sample, the PAM sample was left to settle for 10 minutes so that the beads could disperse evenly.

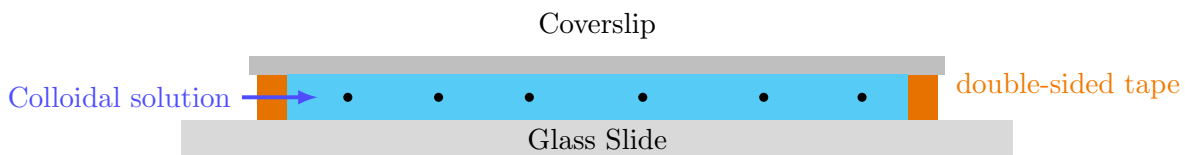


Figure 2: Cross-sectional view of the sample containing 4 μm spheres in a fluid, which is confined by double-sided tape on the glass slide and sealed with a coverslip. The coverslip contacts immersion oil on the objective during measurements. (Not to scale)

2.2.2 Trap stiffness calibration

To determine the trap stiffness κ in deionised water for both traps, ten consecutive videos were taken of the same bead at 650 frames per second (fps). Each video was recorded with an exposure time of 1.5 ms for a duration of 15.4 s at a temperature of $22.2(1)°C$. ImageJ was used to track the bead's centre coordinates in the image plane, for each video the variance of the positions along x and y was calculated. Using the measured variance along each axis, we obtained one trap stiffness value per axis from $\kappa_x = k_B T/\langle x^2 \rangle$ and $\kappa_y = k_B T/\langle y^2 \rangle$. The overall trap stiffness for each trap was then taken as the average of the two axes, given by Equation 1. The same approach was used to determine the trap stiffness of both traps in the complex fluid at a temperature of $20.2(1)°C$. The same

two sets of ten thermal fluctuation videos for each trap were used to calculate the high frequency response for the complex modulus calculation described in section 3.4.

2.2.3 Trap flipping protocol

The trap flipping measurements were performed in both deionised water and the PAM solution to probe the low-frequency response. For each fluid, an isolated bead was first trapped with the steerable trap and moved close to the vicinity of the fixed trap. The fixed and steerable traps were positioned with a centre-centre separation D_0 along the x and y axes such that $D_0 \lesssim 0.8a$, where a is the bead radius, in order to remain within the linear response regime in which the optical potential is approximately harmonic and the restoring force is linear [14].

During the flipping experiment, the bead was alternately confined in the fixed and steerable traps by switching the active beam with a computer programme. Between each flip, the active trap was held constant for a fixed time interval of 25 seconds so that the bead could relax towards the new equilibrium position. The videos were recorded at 650 fps with an exposure time of 1.5 ms for a total of 15 flips. This procedure was repeated for three independent beads in the same sample, giving three separate flipping videos for each fluid. For each video, the bead's position was tracked relative to the centre of the active trap using ImageJ, giving a time series $\mathbf{r}(t)$ for each flip. The individual relaxation trajectories following each switch were extracted and then aligned such that each flip occurred at $t = 0$. These aligned trajectories were then averaged to obtain the mean displacement $\langle \mathbf{r}(t) \rangle$ for each bead. This mean displacement was normalised by the trap separation D_0 using Equation 4 to obtain the normalised mean trajectory $D(t)$ for all three videos of three separate beads in the same sample for both the water and PAM solutions.

After the trajectories were extracted, the high-frequency response from the single trap thermal fluctuations and the low-frequency response from the trap flipping experiments were processed using an in-house built computer programme. The code takes the normalised displacement and the trap calibration data as input and outputs the frequency-dependent storage and loss moduli, $G'(\omega)$ and $G''(\omega)$, by combining the low and high-frequency responses into a single shear modulus $G^*(\omega)$.

3 Results & Discussion

3.1 Trap stiffness calibration

3.1.1 Trap stiffness in deionised water and PAM solution

The trap stiffness in deionised water was determined from ten thermal fluctuation videos of a single bead in each trap. For the fixed trap, we determined that $\kappa = 1.76(6)$ μN/m, while for the steerable trap the stiffness was $\kappa = 6.02(29)$ μN/m. The same procedure was then repeated for the PAM solution to obtain the following trap stiffness of the fixed trap $\kappa = 2.26(87)$ μN/m and 6.77(8) μN/m for the steerable trap. The uncertainties in κ correspond to the standard error on the mean of ten repeated measurements.

3.1.2 Discussion

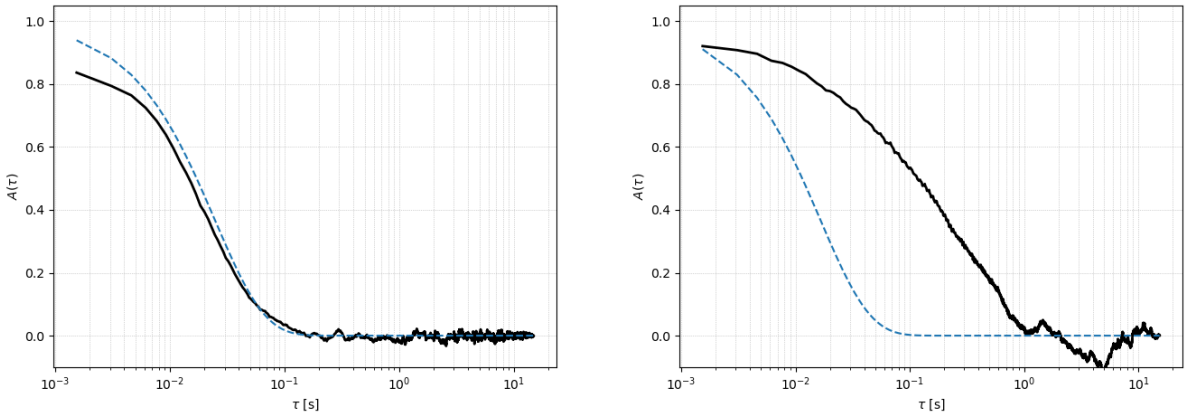
Although not necessary to see interesting results, as seen in later sections, the calibration process was aimed at obtaining both traps to be roughly equal, however, this proved to be difficult in practice. The steerable trap consistently produced a stronger confinement for the same laser current as the fixed trap. Thus, to compensate the fixed trap was set to the maximum current available of 100 mA, whereas the steerable trap was reduced to 65 mA. Even with this reduction, the steerable trap remained stiffer than the fixed one, indicating differences in the optical paths and alignment in both of the beams leading to unequal trapping strengths. The trap stiffnesses measured in both water and PAM solution are broadly consistent across both fluids, in particular the fixed trap increases from $\kappa = 1.76(6)$ μN/m in water to $\kappa = 2.26(87)$ μN/m in the PAM solution, considering the large uncertainty in the latter the results are comparable to each other. The steerable trap remains of similar strength in both fluids, with $\kappa = 6.02(29)$ μN/m in water and 6.77(8) μN/m in the PAM solution, indicating a moderate increase in stiffness. These small differences in trap stiffnesses in both fluids can be attributed to systematic effects such as shifts in height above the coverslip, changes in refractive index and subtle changes in the objective focus. In particular, the beam alignment may have changed as data collection for water and the PAM solution happened on different days. Additionally, drift in the samples may have affected the final values of κ. The addition of drift would contribute to the measured position variance, making $\langle r^2 \rangle$ larger than what it would be for pure thermal fluctuations, since κ is inversely proportional to its variance, such an addition would underestimate the trap stiffness. The close agreement of κ for each trap in water and the PAM solution supports the claim that the trap stiffness is an intrinsic property of the trapping setup, rather than a property of the surrounding fluid.

3.2 High frequency response - single trap fluctuations

The high-frequency response of the fluids is characterised from the same single trap videos used for the stiffness calibration. It is used by computing the normalised autocorrelation function given by Equation 3. The NPAF for deionised water and the PAM solution is shown in Figure 3, with the dashed line representing an ideal single exponential decay for a Newtonian fluid at a temperature of $T = 22°$C. In water, Figure 3a shows that the NPAF decays smoothly with lag time closely resembling the theoretically predicted dashed line over most of the measured range. This indicates that the relaxation is dominated by a single characteristic time, as expected for a Newtonian fluid. Small deviations at the short time scales can be attributed to the finite sampling, while the fluctuations at the longer timescale can be attributed to noise from the finite time series.

In contrast, Figure 3b shows the NPAF measured in the PAM solution. The decay is much slower and is clearly non-exponential in nature, remaining way above the theoretical prediction for a Newtonian fluid, this reflects a broad spectrum of relaxation times. This behaviour shows that the bead experiences a time-dependent viscoelastic drag in the polymer solution, with elastic stresses that relax slower rather than instantaneously like in water, indicating that the relaxation cannot be characterised by a single timescale. The NPAF at the longest lag time again fluctuates around zero, and this behaviour is expected once the true correlations have decayed and similarly to water the estimate becomes dominated by noise from the time series.

These single trap measurements highlight the expected contrast between two fluids. In deionised water the trapped bead relaxes in an almost Newtonian fashion with an approximately single exponential NPAF, while in the PAM solution the bead shows a slower non-exponential decay of the NPAF, indicating a broader spectrum of relaxation times and the presence of viscoelastic behaviour from the surrounding fluid. This difference between water and the PAM solution is consistent with the behaviour reported in [14].



(a) Ionised Water

(b) PAM Solution

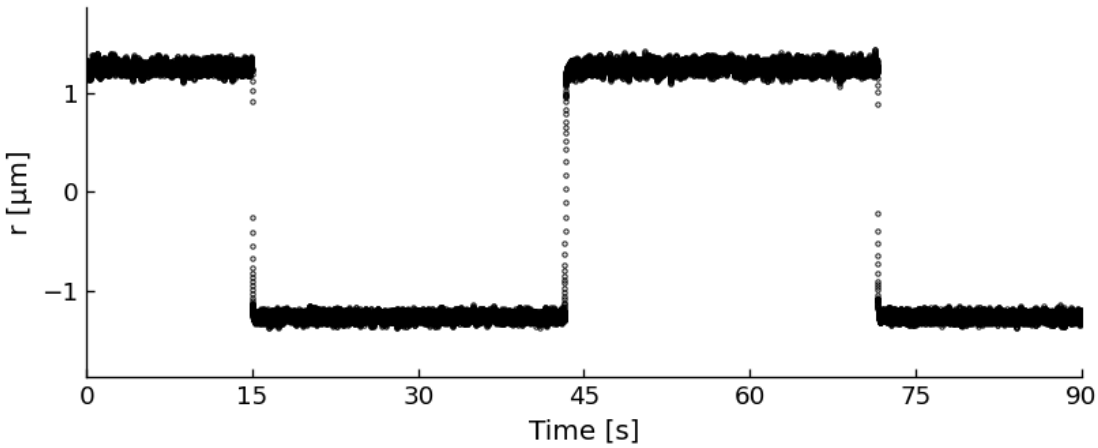
Figure 3: Normalised position autocorrelation function of a trapped 4 μm bead in (a) deionised water and (b) in the PAM solution at $T = 22°$C with trap stiffnesses of $\kappa = 1.76(6)$ μN/m and $\kappa = 2.26(87)$ μN/m respectively. The solid black curves represent how the experimentally measured NPAF from single trap thermal fluctuations compares to the theoretically predicted exponential decay for a Newtonian fluid, shown with the dashed lines.

3.3 Low frequency response - trap flipping

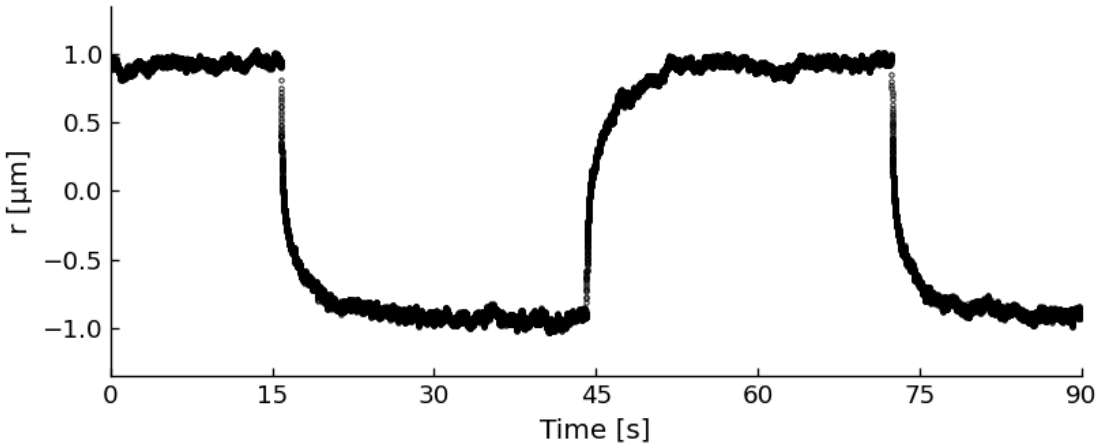
In order to probe the low frequency response of the fluids, the bead was alternately confined in the fixed and steerable traps using the trap flipping protocol. At each trap switch, the active trap was moved by a centre-centre distance and then held in the new trap for 25 seconds, allowing the bead to relax into the new equilibrium position. Figure 4 shows bead trajectories along the flipping axis for a single bead in deionised water and in the PAM solution, and in both Figures 4a and 4b the weaker trap is at the top, while the stronger trap is at the bottom. In both instances, the stronger trap suppresses thermal fluctuations more, leading to a narrower distribution of positions around the equilibrium point. In deionised water (Figure 4a) the bead position switches almost instantaneously between the two traps, and after each switch of the active trap the trajectory of the bead shows a sharp step from one trap to the new active trap. The relaxation back to equilibrium is also almost instantaneous, with only small thermal fluctuations being visible around the mean position of the active traps.

In the PAM solution (Figure 4b) the behaviour differs from that of water. After each flip, the bead does not instantaneously jump to the new equilibrium position, but rather slowly creeps toward it after a rapid movement when the traps are initially switched. A distinct curve like feature is visible rather than a step-like jump, reflecting the gradual

relaxation time of the polymer solution over several seconds, and the bead takes noticeably longer to reach its new equilibrium than in water. Thermal fluctuations of the bead can be observed, which are not as prominent in the PAM solution as they are in water. This highlights the viscoelastic nature of the PAM solution where the elastic stresses built up during the switch slowly relax, so that the bead lags behind the trap and retains memory of its previous position for much longer than in water.



(a) Deionised Water

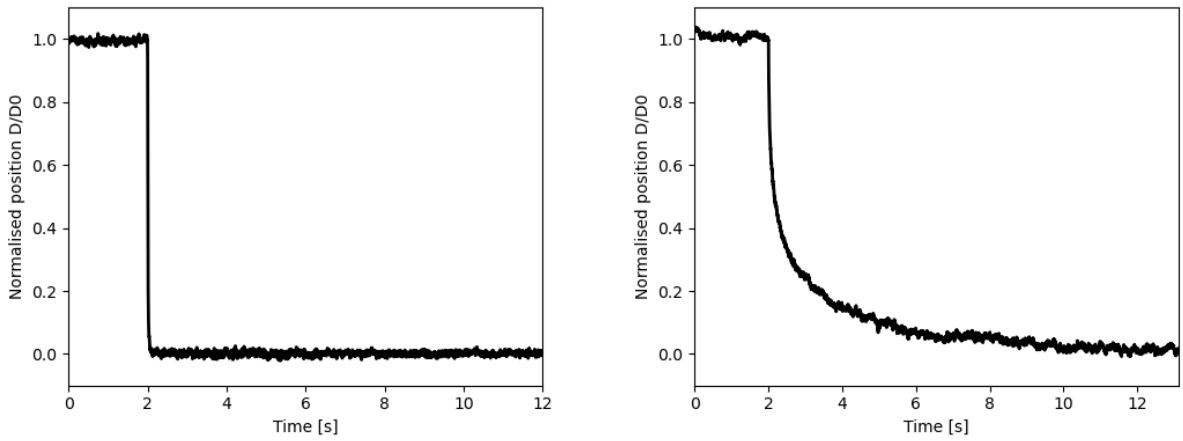


(b) PAM Solution

Figure 4: Position of a 4 μm bead during trap flipping at a temperature of $T = 22°$C. Each panel shows a 90 second snapshot from a 380 second trajectory recording for a single video. In (a) the bead is flipped in deionised water with the bottom trap stiffness of $\kappa = 1.76(6)$ μN/m and the top trap stiffness of $\kappa = 6.02(29)$ μN/m showing an almost instantaneous flip, while (b) shows the same trap configuration (top: $\kappa = 2.26(87)$ μN/m, bottom: 6.77(8) μN/m), where the bead relaxes more slowly towards the new equilibrium position.

The normalised mean displacement $D(t)$ for deionised water and for the PAM solution is shown in Figure 5. This was obtained with the use of Equation 4, by averaging all

flips from three beads in each fluid. Immediately after each switch $D(0) \approx 1$, and the decay towards zero reflects how quickly the bead relaxes towards its new trap centre. Figure 5a shows the normalised displacement in water, which drops almost immediately from 1 to values close to zero, where it fluctuates around a constant mean due to Brownian motion. The almost instantaneous relaxation towards its new equilibrium position is consistent with the behaviour of a bead in a low viscosity Newtonian fluid. The normalised displacement in the PAM solution is shown in Figure 5b, where the relaxation is much slower. Similarly to water, the curve starts off close to 1, but after the initial drop the displacement continues to decrease gradually over several seconds. The shape of the decay is more rounded than the shape of the water decay, indicating that the bead feels a time-dependent resistance from the surrounding polymer solution. This slow, non-instantaneous decay of the normalised displacement $D(t)$ is the low frequency counterpart of the NPAF observed in the high frequency measurements. It reflects the viscoelastic properties of the PAM solution, in particular, the deformation part that is stored elastically and released gradually.



(a) Deionised Water

(b) PAM Solution

Figure 5: Normalised bead displacement $D(t)$ following a trap step down for the (a) deionised water and (b) PAM solution. In (a) the top corresponds to the fixed trap $\kappa = 1.76(6)$ μN/m and bottom corresponds to the steerable trap $\kappa = 6.02(29)$ μN/m, showing an almost instantaneous relaxation to the new equilibrium. In (b) the same trap configuration is used (top: $\kappa = 2.26(87)$ μN/m, bottom: 6.77(8) μN/m), where the decay of $D(t)$ is much slower.

The step-down trajectories as shown in Figure 4 and the normalised displacements, $D(t)$ as shown in Figure 5, convey a clear contrast between the almost instantaneous relaxation in deionised water and the slow, extended relaxation in the PAM solution. In water, the bead reaches its new equilibrium nearly instantaneously and similarly $D(t)$ rapidly decays to a small noise-dominated baseline, whereas in PAM a rapid initial drop is followed by a long, gradual decay that reaches the baseline after several seconds, indicating a broad spectrum of slow relaxation processes in the polymer solution. These results closely match the results obtained in [14], supporting the conclusion that the implementation of these methods successfully captures the expected low frequency viscoelastic

response of a complex fluid.

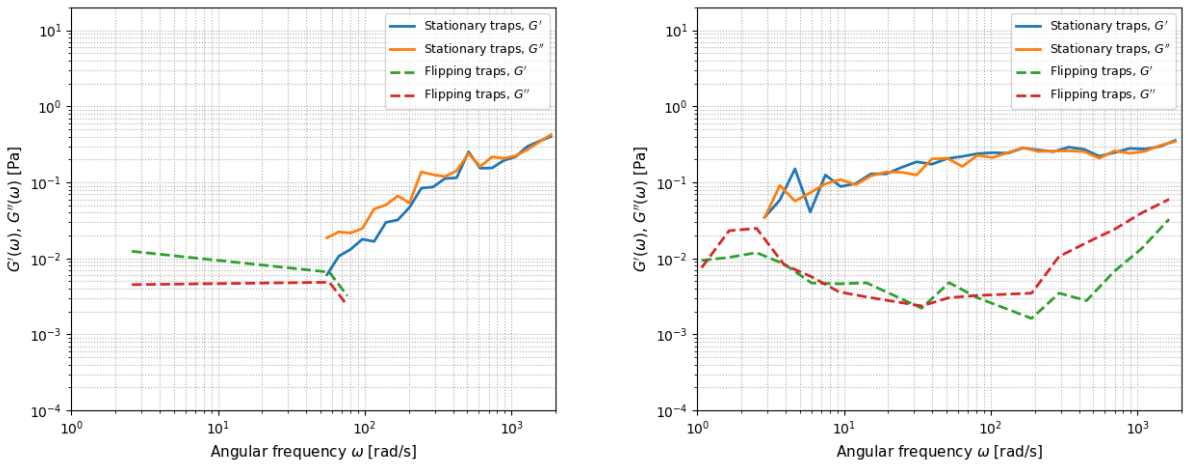
3.4 Viscoelastic moduli

The storage and loss moduli for each fluid were obtained by combining the stationary trap and flipping trap data with the analysis program described in section 2.2.3, using the complex modulus relation in Equation 2. For water, these moduli are shown in Figure 6a. The absolute values of both $G'(\omega)$ and $G''(\omega)$ remain small, as expected for a low viscosity Newtonian fluid. In the high frequency response which corresponds to the stationary traps, the moduli are roughly similar in magnitude, with $G''(\omega)$ being just slightly larger than $G'(\omega)$ over most of the frequency range, and both increase approximately linearly with ω. In the flipping trap data which probes the low frequency response, the measured storage modulus is larger than the loss modulus, and both moduli vary linearly with ω with a drop-off near the higher frequencies, where the stationary traps take over. This behaviour is not what we would expect from a Newtonian liquid at low frequencies, where $G'(\omega) > G''(\omega)$ in the terminal regime, but this is likely a consequence of the combination of the weak bead fluctuations in water being comparable to experimental noise, the uncertainty in the trap calibration process and the limited dynamic range of the moduli in water. In an ideal terminal regime, as cited in literature, $G' \propto \omega^2$ and $G'' \propto \omega$ [14, 15], but in our case the accessible low frequency window does not reflect the ω^2 scaling of $G'(\omega)$ and the ω dependence of $G''(\omega)$ is only weakly suggested.

Similarly to water, the viscoelastic moduli of the PAM solution were obtained by combining data from the stationary trap and flipping trap experiments using the complex modulus relation shown in Equation 2, and the resulting moduli are shown in Figure 6b. In the high frequency response, measured with stationary traps, the loss modulus is roughly the same as the storage modulus and both increase with ω, appearing as almost straight lines on the scale. This indicates a frequency-dependent viscoelastic response in which the elastic and viscous contributions are of comparable magnitude, since the storage and loss moduli remain close together across the frequency range, showing that the PAM solution retains a large part of its elastic component even at short time scales. In the low frequency response, measured with the flipping traps, the storage and loss moduli lie quite close together, and they both vary with ω showing a gentle trend downward when the high frequency response from the stationary traps kicks in. This behaviour suggests that the PAM solution does not reach a clear terminal regime in which $G''(\omega) > G'(\omega)$ and the expected low frequency scaling $G' \propto \omega^2$ and $G'' \propto \omega$ would be observed. No meaningful power law can be extracted for either modulus in this low frequency window. This is likely due to scatter in the data as well as the limited range over which the moduli change, so that the underlying trend with ω is only slightly larger than the experimental noise and uncertainties in the trap calibration. Although $G'(\omega)$ and $G''(\omega)$ are comparable in magnitude over the entire frequency range, this indicates that the PAM solution behaves as a viscoelastic material that retains a significant elastic component even at the longer time scales accessible in this experiment, while the overall increase in $G''(\omega)$ compared to water reflects a larger effective viscosity.

Overall, the water and PAM solution did not reproduce the low frequency behaviour reported in [14]. In their measurements, the extended frequency range and high data quality allowed for a clear observation of the terminal scalings $G' \propto \omega^2$ and $G'' \propto \omega$,

particularly, in the polymer solution. By contrast, in our data the low frequency region obtained from the flipping traps, although covering a comparable range of ω, is noisy and flat with no convincing ω or ω^2 dependence that can be identified. In this regard, the combined stationary and flipping trap measurements are not fully successful in the same way as Preece et al. [14], as we cannot extract reliable scaling behaviour or determine how the viscosity behaves in the long time frequency limit. Nevertheless, the plots still contain useful information. For water, the moduli remain small and consistent with an almost Newtonian response, and for the PAM solution, both $G'(\omega)$ and $G''(\omega)$ are much larger and stay in comparable magnitude over the whole accessible range, which demonstrates that the polymer solution is viscoelastic and retains a significant elastic component even at the longest time scales when probed, even though the ideal terminal scaling laws cannot be resolved.



(a) Deionised Water

(b) PAM solution

Figure 6: Storage modulus $G'(\omega)$ and loss $G''(\omega)$ as a function of angular frequency ω for (a) deionised water and (b) the PAM solution. For each liquid, solid lines represent the high frequency response obtained from the stationary trap thermal fluctuations, while the low frequency response shown in dashed lines is obtained from trap flipping experiments.

4 Conclusion

This project has shown that a dual-trap optical tweezer setup can reproducibly characterise both the trapping properties and the viscoelastic response of a simple Newtonian fluid and a polymer solution. From equipartition analysis of thermal fluctuations, the fixed and steerable traps in deionised water were found to have $\kappa = 1.76(6)$ μN/m and $\kappa = 6.02(29)$ μN/m, respectively, while in the PAM solution the corresponding values were $\kappa = 2.26(87)$ μN/m and $6.77(8)$ μN/m. Within uncertainties and experimental procedure, each trap stiffness is therefore essentially unchanged between the two fluids, confirming that the trap stiffness κ is an intrinsic property of the optical trapping setup, providing a solid basis for comparing fluid responses.

The time-dependent measurements consistently highlight a strong contrast between water and the PAM solution. The normalised position autocorrelation function in water

decays rapidly and is well described by an almost single-exponential form, closely following the theoretical Newtonian prediction. In the PAM solution, the NPAF decays much more slowly and is clearly non-exponential, remaining well above the Newtonian curve, indicating a broad distribution of relaxation times and the time-dependent viscoelastic drag. The low frequency trap flipping measurements show the same behaviour in the time domain, with the normalised displacement $D(t)$ for water dropping from $D(t) \approx 1$ to a noise-dominated baseline nearly instantaneously, while in the PAM solution the initial drop is followed by a gradual decay over several seconds before approaching the baseline. These results highlight that the polymer solution stores part of the deformation elastically and releases it slowly, while water relaxes almost instantaneously.

The analysis of the storage and loss moduli from the combined stationary and trap flipping data provides an overview of the contrast across the full frequency range. In water, $G'(\omega)$ and $G''(\omega)$ remain of order 10^{-1}-10^{0} Pa across the measured frequencies, which is consistent with a low-viscosity fluid, although the expected viscous dominance $G''(\omega) > G'(\omega)$ is only seen at high frequencies and in the low-frequency range this behaviour is obscured by noise and calibration uncertainties, failing to show the expected terminal behaviour as reported in Preece et al. [14]. Similarly, for the PAM solution, both moduli remain of order 10^{-1}-10^{0} Pa across the measured spectrum of frequencies, are larger than in water and remain of comparable magnitude over the entire frequency range, demonstrating a viscoelastic response. However, just as for water, the low-frequency response is relatively flat and noisy, so the ideal terminal behaviour of $G' \propto \omega^2$ and $G'' \propto \omega$ as described in [14, 15] could not be observed. Nevertheless, the information provided by the modulus spectra is useful, and the PAM solution shows systematically higher values of both moduli when compared to water. This indicates that the polymer solution is more viscoelastic than water and retains an elastic component over a longer time period.

Overall, the combination of trap stiffness calibration, NPAF analysis, normalised displacement and complex modulus demonstrates that the implemented protocol successfully distinguishes a simple Newtonian fluid from a viscoelastic solution and qualitatively reproduces the behaviour seen in other studies [14]. At the same time, the limitations of the low-frequency modulus data emphasize a need for improved signal-to-noise, better drift control and longer time series if the quantitative terminal behaviour is to be obtained with this optical tweezers approach.

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